

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO4 ASSESSMENT TOOL 1 – Test Case Design Assignment

Course Code:	CSA1003	Course Title:	Software Engineering
CO Assessed:	CO4	Assessment:	Assessment Tool 1 – Test Case Design Assignment
Weightage:	20%	Total Marks:	25
CO4 Statement:	Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics		
Student Name:	ALIGETI AKSHAYA	Reg. No.:	192425170
Date:	20-08-2026	Duration:	60 minutes

Code of Conduct

I A.Akshaya(192425170) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Akshaya

Annexure A – Test Case Design Assignment

Instructions to Students:

ACADEMIC PROJECT, ALLOCATION MONITORING AND EVALUATION MANAGEMENT SYSTEM

Your test case design should include:

1. Identify the **functional and non-functional requirements** to be tested.

Functional Requirements

The following functional requirements should be tested:

1. Student registration and login.
2. Faculty, HOD, Project Coordinator, and Admin login.
3. Student profile creation and updating.
4. Selection of student interests.
5. Selection of programming languages and technical skills.
6. Project creation and management.
7. Interest-based project allocation.

8. Faculty guide assignment.
9. Project progress submission.
10. Faculty progress monitoring.
11. Project deadline management.
12. Internal and external evaluation.
13. Marks and feedback submission.
14. Role-based access control.
15. Search and filtering of project information.
16. Report generation.

Non-Functional Requirements

The following quality attributes should also be tested:

- **Performance** – Pages and requests should respond within an acceptable time.
- **Security** – Users should access only the features permitted for their roles.
- **Usability** – The system should be simple and easy to understand.
- **Reliability** – The system should work correctly without frequent failures.
- **Availability** – The system should be available when required.
- **Scalability** – The system should support an increasing number of students and projects.
- **Maintainability** – The code should be easy to modify and maintain.

2. Identify the **test scenarios** for the assigned problem statement.

Scenario ID	Test Scenario
TS01	Student registration
TS02	User login
TS03	Student profile management
TS04	Student interest and skill selection
TS05	Project creation
TS06	Project allocation
TS07	Faculty guide assignment
TS08	Project progress submission
TS09	Faculty progress monitoring
TS10	Deadline management
TS11	Evaluation and marks

Scenario ID	Test Scenario
TS12	Feedback management
TS13	Role-based access
TS14	Search and filtering
TS15	Report generation
TS16	System performance
TS17	Security validation
TS18	Database reliability

3. Prepare detailed **test cases** covering normal, boundary, invalid, and exceptional conditions.

I D	Test Scena rio	Test Steps	Test Data	Expec ted Result	Act ual Res ult	St at us
T C 0 1	Studen t Regist ration	Open registration → Enter details → Submit	Valid name, email, roll no., password	Studen t accou nt create d	As exp ecte d	Pa ss
T C 0 2	Regist ration with missin g data	Leave required field empty → Submit	Email = blank	Valida tion messa ge displa yed	As exp ecte d	Pa ss
T C 0 3	Duplic ate registr ation	Register using existing email	Existing email	Duplic ate accou nt should not be create d	As exp ecte d	Pa ss
T C 0 4	Valid Login	Enter valid credentials → Login	Valid email/pa ssword	User dashb oard opens	As exp ecte d	Pa ss
T C 0 5	Invali d Login	Enter incorrect password	Wrong password	Error messa ge	As exp ecte d	Pa ss

I D	Test Scena rio	Test Steps	Test Data	Expec ted Result	Act ual Res ult	St at us
				displa yed		
T C 0 6	Studen t Profile	Login → Open profile → Enter details → Save	Departm ent, skills, interests	Profile succes sfully saved	As exp ecte d	Pa ss
T C 0 7	Interes t Selecti on	Select interests → Save	AI, Web Develop ment	Select ed interes ts stored	As exp ecte d	Pa ss
T C 0 8	Skill Selecti on	Select programmin g languages	Java, Python	Skills succes sfully saved	As exp ecte d	Pa ss
T C 0 9	Projec t Creati on	Coordinator → Add project → Submit	Project name, domain, skills	Projec t create d	As exp ecte d	Pa ss
T C 1 0	Projec t Alloca tion	Select student → Select suitable project → Allocate	Student + Web Develop ment project	Projec t allocat ed succes sfully	As exp ecte d	Pa ss
T C 1 1	Duplic ate Alloca tion	Allocate another project to same student	Existing allocated student	Syste m should preven t invalid duplic ate allocat ion	As exp ecte d	Pa ss
T C 1 2	Facult y Assign ment	Coordinator assigns faculty → Save	Faculty + Project	Facult y assign ed succes sfully	As exp ecte d	Pa ss

I D	Test Scena rio	Test Steps	Test Data	Expec ted Result	Act ual Res ult	St at us
T C 1 3	Progre ss Submi ssion	Student → Progress → Submit	Weekly progress report	Report stored succes sfully	As exp ecte d	Pa ss
T C 1 4	Empty Progre ss Report	Submit without entering report	Blank report	Valida tion messa ge displa yed	As exp ecte d	Pa ss
T C 1 5	Facult y Monit oring	Faculty → Dashboard	Assigned student	Studen t progre ss displa yed	As exp ecte d	Pa ss
T C 1 6	Deadli ne Check	Set project deadline	Valid future date	Deadli ne succes sfully stored	As exp ecte d	Pa ss
T C 1 7	Invali d Deadli ne	Enter past date	Past date	Syste m rejects the date	As exp ecte d	Pa ss
T C 1 8	Evalua tion Marks	Faculty → Enter marks → Submit	Marks = 85	Marks saved succes sfully	As exp ecte d	Pa ss
T C 1 9	Bound ary Marks	Enter minimum/m aximum marks	0 and 100	Values accept ed	As exp ecte d	Pa ss
T C 2 0	Invali d Marks	Enter marks outside range	-5 or 105	Invali d marks rejecte d	As exp ecte d	Pa ss
T C	Feedb ack	Faculty enters	"Good progress"	Feedb ack saved	As exp	Pa ss

ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
21		feedback → Submit		and visible to student	ected	
TC22	Role Access	Login as student → Open Admin page	Student account	Access denied	As expected	Pass
TC23	Search Project	Enter project keyword → Search	"AI"	Matching projects displayed	As expected	Pass
TC24	Report Generation	Select project → Generate report	Valid project	Report generated successfully	As expected	Pass
TC25	Database Failure	Stop database → Try login	Valid credentials	System displays appropriate error instead of crashing	As expected	Pass

4. Include **Test Case ID**, **Test Scenario**, **Test Steps**, **Test Data**, **Expected Result**, **Actual Result**, and **Pass/Fail Status**.

Each test case contains the required fields:

- **Test Case ID** – Unique identification number.
- **Test Scenario** – Functionality being tested.
- **Test Steps** – Actions performed by the tester.

- **Test Data** – Input values used during testing.
 - **Expected Result** – What should happen.
 - **Actual Result** – What actually happened during execution.
 - **Pass/Fail Status** – Whether the test case passed or failed.
5. Apply appropriate **test design techniques** such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, or State Transition Testing wherever applicable.

A. Equivalence Partitioning

Equivalence Partitioning divides input data into valid and invalid groups.

Example: Evaluation Marks

Valid marks:

0 – 100

Invalid groups:

Less than 0

More than 100

Alphabetic values

Special characters

Input	Partition	Expected
50	Valid	Accept
0	Boundary valid	Accept
100	Boundary valid	Accept
-1	Invalid	Reject
101	Invalid	Reject
ABC	Invalid	Reject

B. Boundary Value Analysis

Boundary values are tested because errors commonly occur at the limits.

For **evaluation marks**, the boundary values are:

-1 0 1

99 100 101

Expected:

- 0 → Accept
- 100 → Accept
- -1 → Reject
- 101 → Reject

C. Decision Table Testing

Decision table testing can be applied to **project allocation**.

Student Interest	Required Skill	Project Available	Allocation
Yes	Yes	Yes	Allocate
Yes	Yes	No	Do not allocate
Yes	No	Yes	Do not allocate
No	Yes	Yes	Do not allocate
No	No	No	Do not allocate

D. State Transition Testing

State Transition Testing can be applied to the **project lifecycle**.

Project Created



Project Allocated



Project In Progress



Progress Submitted



Faculty Review



Evaluation



Project Completed

6. Ensure that the test cases provide adequate **requirement and functional coverage** for the assigned system.

Requirement	Covered Test Cases
Registration	TC01–TC03
Login	TC04–TC05
Student Profile	TC06
Interests & Skills	TC07–TC08
Project Management	TC09
Project Allocation	TC10–TC11
Faculty Assignment	TC12
Progress Tracking	TC13–TC15
Deadline Management	TC16–TC17
Evaluation	TC18–TC20
Feedback	TC21
Security / Role Access	TC22
Search	TC23
Reports	TC24
Reliability	TC25

Deliverable: Submit a Test Case Design document containing the identified scenarios, test cases, test data, expected results, and test coverage based on the individual problem statement.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Requirement & Test Scenario Identification(15%)	Clearly identifies all relevant functional/non-functional requirements and comprehensive test scenarios	Identifies most requirements and scenarios	Identifies basic requirements and scenarios	Requirements/scenarios are incomplete or unclear

Test Case Design & Completeness(25%)	Comprehensive test cases covering normal, boundary, invalid, and exceptional conditions	Covers most important conditions with minor gaps	Covers basic conditions but misses several important cases	Test cases are very limited or incomplete
Test Data & Expected Results(20%)	Appropriate test data with precise, measurable expected results for every test case	Mostly appropriate test data and expected results	Some test data/results are unclear or incomplete	Test data and expected results are largely missing/incorrect
Application of Test Design Techniques(15%)	Correctly applies multiple techniques such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, and State Transition	Correctly applies at least two relevant techniques	Applies one technique with limited justification	Techniques are not applied or incorrectly applied
Traceability & Test Coverage(15%)	Excellent mapping between requirements, scenarios, and test cases with strong coverage	Good traceability and coverage with minor gaps	Partial traceability and moderate coverage	Poor/no traceability and inadequate coverage
Documentation & Presentation(10%)	Well-structured, clear, professional, consistent, and easy to understand	Well-organized with minor presentation issues	Adequately organized but has some clarity/formatting issues	Poorly organized, unclear, or incomplete documentation

Criteria	Mark
Requirement & Test Scenario Identification (15%)	/5
Test Case Design & Completeness (25%)	/4
Test Data & Expected Results (20%)	/4
Application of Test Design Techniques (15%)	/4
Traceability & Test Coverage (15%)	/4
Documentation & Presentation (10%)	/4
TOTAL	/25

